Abstract

This thesis presents the development and characterization of the experimental system required for the preparation, optical trapping, and controlled transport of cold ^{87}Rb atoms toward a Hollow-Core Photonic Crystal Fiber (HCPCF).
First, the atomic sample preparation was optimized: the optical molasses reached a minimum temperature of $T = (8.4 \pm 0.2)\,\mu$K, though the ODT-loading configuration differed by a few tenths of a μK. Loading into the Optical Dipole Trap (ODT) yielded a loading efficiency of $(0.161 \pm 0.005)\,\%$ and a conservatively estimated trapped population of $(1.0 \pm 0.1) \times 10^5$ atoms.
To implement the optical conveyor belt, a second counter-propagating 1064 nm field was established using a polarization-maintaining single-mode fiber. Characterization of the transmitted field showed fractional power and angular Allan deviations reaching minima of 8×10^{-4} at 0.1 s and 7×10^{-5} rad at 3 s, respectively, before degrading at longer times from low-frequency drifts.
To counteract potential heating during optical transport, Electromagnetically Induced Transparency (EIT) cooling was investigated numerically via a quantum master-equation model in QuTiP. Multilevel simulations of the ^{87}Rb D_2 transition, with an active repump field preventing optical pumping into uncoupled states, reached a final $\langle n \rangle \approx 0.08$ in approximately 18 ms.
These results establish the main experimental and numerical prerequisites for a future implementation of conveyor-belt transport and sub-Doppler cooling of cold atoms inside the HCPCF.